

# ConstructionSafety-FaithBench: Counterfactual Evaluation of Visual-Evidence Faithfulness in Vision-Language Safety Inspection

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## Abstract

Construction-safety vision-language models are often evaluated by final answers or plausible rationales, but neither shows whether a model used the rule-relevant visual evidence. We present ConstructionSafety-FaithBench, a compact benchmark and counterfactual audit pipeline for construction-safety VLMs. Each example pairs a site image with a normalized safety rule, an answer label, and localizable visual evidence. The implemented artifact includes 163 pilot image-rule pairs, 588 scale-up candidates, four safety-rule families, normalized model-output schemas, deterministic baselines, a Florence-2 grounding adapter, 948 targeted/matched-random occlusion runs, and a 120-row final audit-label layer. On the audit set, the current Florence grounding adapter reaches only 18.3% answer accuracy, while a metadata-assisted bootstrap reaches 78.3%, showing that the prioritized audit split exposes substantial model weakness hidden by scaffold-level validation. Targeted evidence occlusion changes Florence answers 39.2% of the time versus 9.2% for matched random masks, with a paired answer-flip gap of 30.0 points. The current final-label layer combines two independent AI-generated audit passes with returned adjudication for all 12 disagreements; we therefore report it with explicit provenance rather than as unqualified human ground truth.

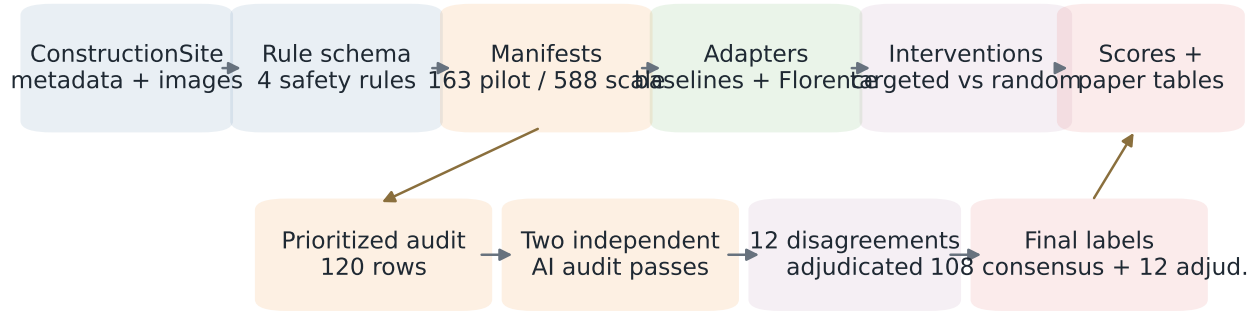
## 1 Introduction

Vision-language models (VLMs) could help screen construction imagery for safety-rule violations, but answer accuracy alone is too weak for this setting. A model may answer “violation” while grounding its decision on background scaffolding instead of a missing hard hat, or it may keep the same answer after the rule-relevant worker is masked. In safety inspection, this distinction matters: explanations are useful only when the answer, evidence, and intervention behavior agree.

ConstructionSafety-FaithBench operationalizes this distinction as a compact, reproducible audit suite. Each row binds an image, a natural-language safety question, a normalized rule, an answer label, and evidence boxes. Models must output both an answer and evidence regions. We then compare ordinary answer scores with evidence overlap and counterfactual sensitivity under targeted and same-size random masks.

The current implementation is intentionally small but end-to-end. It includes a 163-row pilot manifest, a 588-row scale-up candidate split, a 120-row prioritized audit set, deterministic baselines, a local Florence-2 grounding adapter, and a completed targeted-versus-random occlusion study. The final audit-label layer combines 108 A/B consensus rows with 12 returned adjudication decisions. Because the two audit passes were AI-generated, we preserve the provenance explicitly and avoid claiming unqualified human ground truth.

**Contributions.** This repository contributes: (i) a normalized four-rule construction-safety benchmark schema; (ii) pilot, scale-up, audit, intervention, and final-label artifacts with validation scripts; (iii) a model-output and scoring interface that separates answer correctness, evidence overlap, and



All outputs use normalized JSONL/CSV schemas and SHA-256 release manifests.

Figure 1: ConstructionSafety-FaithBench pipeline. Source metadata and images are normalized into rule-specific manifests, scored through common model-output schemas, audited through two-pass/adjudicated final labels, and evaluated with answer, evidence, and intervention metrics.

counterfactual faithfulness; (iv) a 948-run Florence-2 occlusion study showing larger changes under rule-evidence masking than matched random masking; and (v) a compact paper/reproducibility package with generated figures, tables, checks, and SHA-256 artifact manifests.

## 2 Related Work

**Construction-safety VLMs.** Construction-safety VLM work has begun to evaluate hazard identification, rule-based VQA, captioning, and grounding. ConstructionSite is the dataset foundation for this study because it links site imagery with captions, rule-violation records, and grounding metadata [1]. We do not introduce a new detector; instead, we ask whether a VLM-style safety inspector uses the correct visual evidence when answering rule questions. Florence-2 is the first local image-grounded adapter in this repository because it supports prompted grounding and detection-style outputs [4].

**Faithfulness and rationales.** Prior XAI work shows that plausible explanations are not necessarily faithful; attention and rationales can be persuasive without being causal [2]. This motivates evaluating answer labels, evidence boxes, and counterfactual response separately. Perturbation tests are especially relevant for safety imagery because a faithful model should react more to masking a missing hard hat, unprotected edge, or worker-equipment proximity cue than to a same-size mask elsewhere.

**Evaluation and datasets.** The paper is framed as an evaluation artifact: benchmark schema, audit labels, model-output normalization, intervention protocol, score tables, and failure slices. This aligns with top-tier evaluation/dataset contributions rather than new-model claims. The safety framing follows risk-management guidance emphasizing documented scope, measurement limits, provenance, and prohibited high-stakes use [3].

### 3 Benchmark

Each benchmark example consists of an image identifier, source split, normalized safety rule, natural-language question, answer label, evidence metadata, and provenance fields. The four implemented rule families are hard-hat/PPE compliance, harness/fall protection, guardrail/edge protection, and struck-by/equipment proximity. The benchmark stores image identifiers and derived metadata, but does not redistribute raw source images in the repository.

The pilot manifest contains 163 image-rule pairs from the ConstructionSite test split. The scale-up candidate manifest contains 588 test-split image-rule pairs selected from source metadata. Table 1 shows the scale-up class counts; the split remains imbalanced because struck-by and fall-hazard source cases are scarce under the current priority-labeling policy.

**Annotation layers.** We maintain three label layers. First, model-assisted bootstrap annotations are generated from source metadata, captions, rule-violation records, and available boxes. They are useful for scaling and triage but are not ground truth. Second, `human_audit_batch_001` is a balanced 120-row audit queue with 30 rows per rule, prioritized for Florence/bootstrap disagreement and weak evidence overlap. Third, the final audit-label layer resolves the audit queue: 108 rows use A/B answer agreement from two independent AI-generated passes, and the 12 A/B disagreements use the returned adjudication package. This produces 83 violation, 31 compliant, and 6 uncertain labels. We report this layer as adjudicated audit labels with AI-pass provenance.

Table 1: Scale-up candidate manifest class counts.

| Class          | Count |
|----------------|-------|
| Compliant      | 250   |
| Fall Hazard    | 75    |
| Ppe Violation  | 250   |
| Struck By Risk | 13    |
| Total          | 588   |

**Architecture.** Figure 1 shows the complete artifact flow. Manifests and annotations feed normalized model adapters. Model outputs are scored against either bootstrap labels or final audit labels. Intervention specs reuse the same evidence boxes to compare targeted masks with same-size random controls. Every generated table is reproducible from checked CSV/JSONL files and is covered by lightweight validation.

### 4 Metrics

The benchmark separates *what* a model answers from *where* it looks and *whether* that evidence affects the answer.

**Answer metrics.** For a reference label  $y_i$  and normalized prediction  $\hat{y}_i$ , we report accuracy, macro-F1 over `compliant`, `violation`, and `uncertain`, invalid-output rate, and nonambiguous-example accuracy. Bootstrap-label scores measure pipeline consistency; final-audit scores measure behavior against the completed 120-row audit layer.

**Evidence metrics.** For predicted boxes  $\hat{B}$  and reference boxes  $B$ , we report evidence presence, best IoU, normalized centroid drift, and evidence disappearance. IoU alone is insufficient because small PPE regions can produce large IoU swings from tiny localization errors; centroid drift and disappearance expose evidence relocation even when overlap is hard to interpret.

**Intervention metrics.** Each row with a target box receives one rule-evidence occlusion and five same-size matched-random occlusions. The key paired statistic is  $\Delta_{\text{target-random}}$ : the targeted answer-flip or evidence-drift effect minus the per-example matched-random mean. A faithful evidence-using model should change more under targeted masks than under matched random masks, but should not collapse to arbitrary evidence elsewhere.

## 5 Experimental Setup

The experiment suite has three parts: answer scoring, evidence scoring, and counterfactual intervention. All adapters write the same normalized fields: answer label, evidence regions, invalid-output flag, and provenance metadata.

- **Annotation bootstrap:** echoes the model-assisted annotation labels and evidence.
- **Florence grounding:** runs the local Florence-2 grounding pipeline from the companion XAI pilot and maps grounded detections through a deterministic geometric decision layer.
- **Manifest seed:** echoes the pilot weak label from the manifest.
- **Majority violation:** image-blind baseline that always predicts violation.
- **Caption keyword:** caption-only leakage baseline using rule-specific keywords and no image pixels.

We evaluate these adapters on the pilot manifest, scale-up candidate manifest, and 120-row final audit-label layer. The intervention study is run for Florence on the pilot set because it is the first local image-grounded adapter with box-like outputs. The frozen intervention file contains 158 targeted masks and 790 matched-random controls. The current next empirical extension is to add instruction-following VLM adapters behind the same schema, record model version and decoding settings, and rerun both answer scoring and intervention scoring for the strongest model.

## 6 Results

**Final audit-label layer.** The 120-row audit set is intentionally difficult: it over-samples Florence/bootstrap disagreements and weak evidence overlap. Figure 2 summarizes the label distribution and answer scores. The final labels contain 83 violation, 31 compliant, and 6 uncertain rows. Florence grounding reaches only 18.3% accuracy and 12.5% macro-F1, while the metadata-assisted bootstrap reaches 78.3% accuracy and 50.5% macro-F1. This gap is useful: the bootstrap layer remains a scaling aid, but the audit layer exposes visually grounded failure modes that bootstrap validation alone would miss.

**Bootstrap validation.** Table 3 reports pilot scores against model-assisted bootstrap annotations. These numbers validate the harness and make baselines comparable, but they are not final benchmark claims. The annotation-bootstrap row is an upper-bound sanity check because it echoes the reference layer.

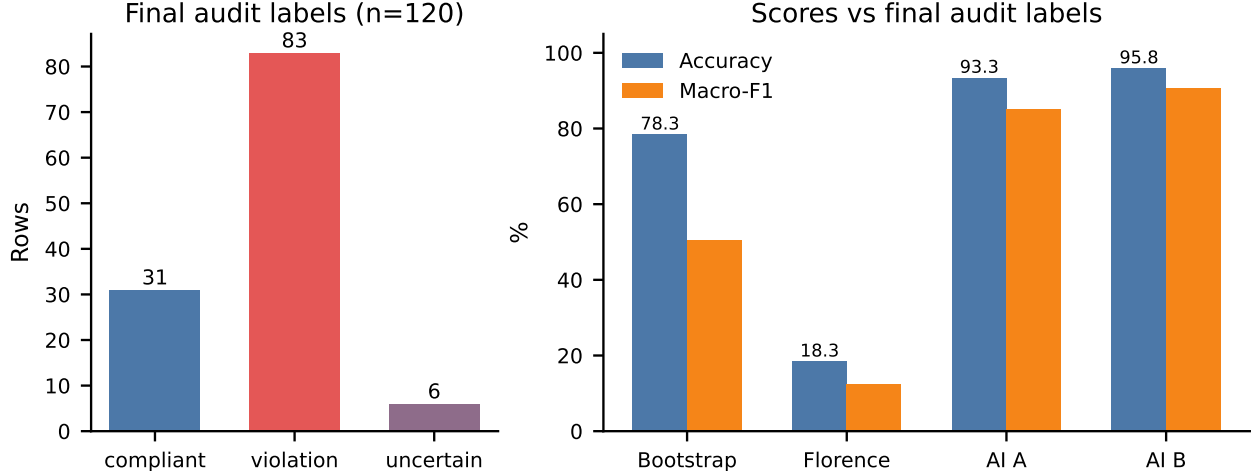


Figure 2: Audit-final label distribution and answer scores. The final layer combines 108 A/B consensus labels with 12 returned adjudication decisions; it is complete but reported with explicit AI-pass provenance.

Table 2: Audit-final answer scores against the 120-row final-label layer.

| Source               | N   | Accuracy | Macro-F1 |
|----------------------|-----|----------|----------|
| Annotation bootstrap | 120 | 78.3%    | 50.5%    |
| Florence grounding   | 120 | 18.3%    | 12.5%    |
| AI annotator 1       | 120 | 93.3%    | 85.2%    |
| AI annotator 2       | 120 | 95.8%    | 90.5%    |

**Counterfactual faithfulness.** Table 4 summarizes the frozen pilot intervention specifications. Figure 3 and Table 5 show that targeted rule-evidence masks affect Florence much more than matched random masks. Targeted occlusion flips answers in 39.2% of rows versus 9.2% for random controls; the paired answer-flip gap is 30.0 points with a bootstrap 95% interval of 22.3 to 37.7 points. Targeted masks also cause much larger centroid drift, indicating evidence relocation rather than only answer instability.

**Scale-up diagnostics.** Table 6 reports adapters on the 588-row scale-up candidate manifest against bootstrap annotations. The caption-keyword baseline is a leakage diagnostic; its poor performance indicates that this simple caption-only heuristic cannot solve the split. Figure 4 shows that Florence usually emits some evidence, yet answer accuracy remains weak, especially for PPE hard-hat and struck-by equipment rules. This separation between evidence presence and answer correctness is the central empirical reason for reporting answer, grounding, and counterfactual metrics separately.

## 7 Limitations and Ethics

The study is a compact audit artifact, not a complete field-deployment benchmark. First, the 120-row final-label layer is complete but has explicit AI-pass provenance: two independent AI-generated passes plus returned adjudication for disagreements. It should be used as an adjudicated audit layer,

Table 3: Pilot baseline scores against model-assisted bootstrap annotations.

| Adapter              | Accuracy | Macro-F1 | Evidence present | Nonambig. acc. |
|----------------------|----------|----------|------------------|----------------|
| Annotation bootstrap | 100.0%   | 100.0%   | 98.8%            | 100.0%         |
| Florence grounding   | 41.7%    | 41.7%    | 96.9%            | 37.7%          |
| Manifest seed        | 100.0%   | 100.0%   | 0.0%             | 100.0%         |
| Majority violation   | 69.3%    | 40.9%    | 0.0%             | 72.5%          |

Table 4: Frozen pilot intervention specifications.

| Intervention type        | Count |
|--------------------------|-------|
| Targeted Occlusion       | 158   |
| Matched Random Occlusion | 790   |
| Total                    | 948   |

not described as unqualified human ground truth. Second, the only current image-grounded model is a Florence-2 grounding adapter with deterministic decision logic; stronger instruction-following VLMs and closed-model adapters must be added before broad model-ranking claims. Third, the scale-up split remains imbalanced because struck-by and fall-hazard source examples are scarce.

Construction-site imagery raises privacy, surveillance, and labor-governance concerns. This benchmark must not be used for autonomous inspection, worker discipline, or surveillance. Appropriate use is research evaluation of VLM faithfulness under documented scope limits. Future releases should preserve source licensing, avoid redistributing images when rights are unclear, document non-commercial restrictions, include prohibited-use language, and require human safety professionals for any real inspection workflow.

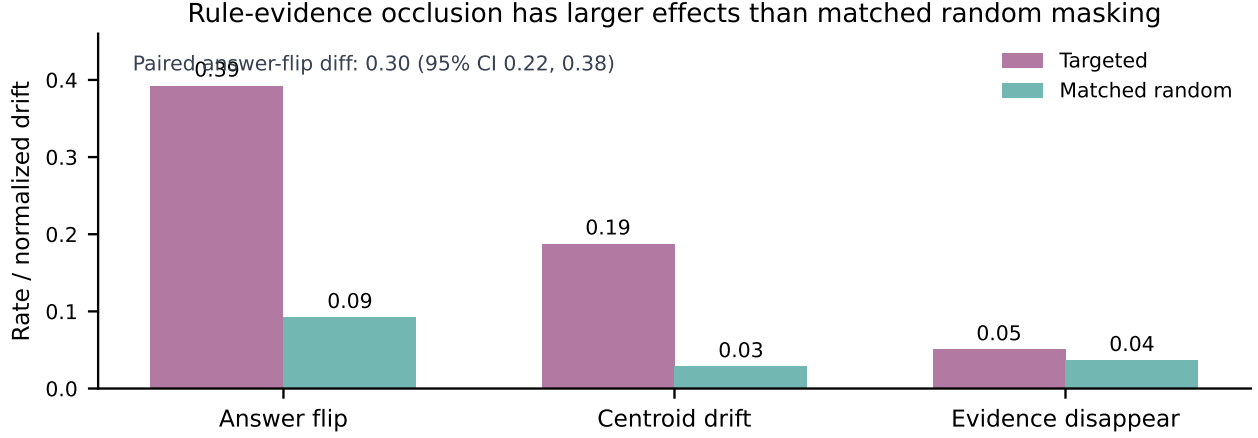


Figure 3: Florence intervention behavior. Rule-evidence masks induce more answer flips and evidence drift than matched random masks, supporting the need for counterfactual faithfulness metrics beyond answer accuracy.

Table 5: Pilot Florence intervention effects.

| Metric                 | Targeted | Matched random | Paired diff. |
|------------------------|----------|----------------|--------------|
| Answer flip rate       | 39.2%    | 9.2%           | 30.0%        |
| Centroid drift         | 0.188    | 0.029          | 0.160        |
| Evidence disappearance | 5.1%     | 3.7%           | –            |

Table 6: Scale-up baseline scores against model-assisted bootstrap annotations.

| Adapter              | Accuracy | Macro-F1 | Evidence present | Nonambig. acc. |
|----------------------|----------|----------|------------------|----------------|
| Annotation bootstrap | 100.0%   | 100.0%   | 57.3%            | 100.0%         |
| Florence grounding   | 39.3%    | 38.6%    | 98.8%            | 28.3%          |
| Manifest seed        | 100.0%   | 100.0%   | 0.0%             | 100.0%         |
| Majority violation   | 57.5%    | 36.5%    | 0.0%             | 82.7%          |
| Caption keyword      | 19.7%    | 26.6%    | 0.0%             | 8.1%           |

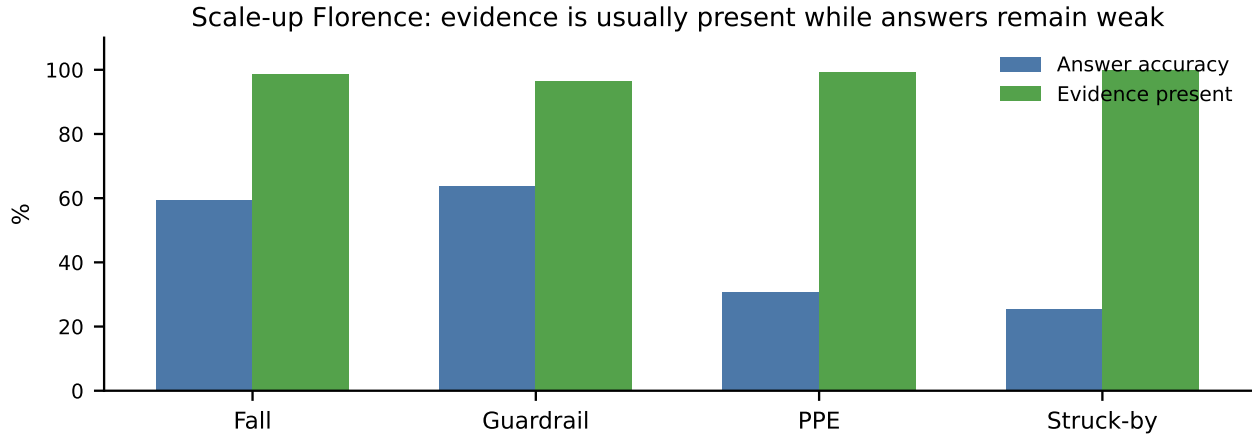


Figure 4: Scale-up Florence rule slices. Evidence regions are usually present, but answer accuracy remains rule-dependent and weak on PPE and struck-by examples.

Table 7: Scale-up Florence rule slices.

| Rule                | N   | Accuracy | Evidence present | Mean IoU |
|---------------------|-----|----------|------------------|----------|
| Fall harness        | 69  | 59.4%    | 98.6%            | 0.130    |
| Guardrail edge      | 110 | 63.6%    | 96.4%            | 0.171    |
| PPE hard hat        | 303 | 30.7%    | 99.3%            | 0.030    |
| Struck-by equipment | 106 | 25.5%    | 100.0%           | 0.033    |

## References

- [1] X. Chen and Z. Zou. Are large pre-trained vision language models effective construction safety inspectors?, 2025.
- [2] Sarthak Jain and Byron C. Wallace. Attention is not explanation. *Proceedings of NAACL-HLT*, 2019.
- [3] National Institute of Standards and Technology. Artificial intelligence risk management framework (ai rmf 1.0), 2023.
- [4] Bin Xiao, Haiping Wu, Weijian Xu, Xiyang Dai, Houdong Hu, Yumao Lu, Michael Zeng, Ce Liu, and Lu Yuan. Florence-2: Advancing a unified representation for a variety of vision tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

## A Detailed Results and Reproducibility Status

This appendix records the concrete artifacts, counts, and result tables currently implemented in the repository. All numbers in this appendix are generated from checked artifacts. The model-assisted annotation layer remains a bootstrap reference for pipeline validation and audit prioritization; the 120-row final audit-label layer is complete but must be reported with its AI-pass and returned-adjudication provenance.

### A.1 Implemented Artifact Set

The current scaffold contains the following reproducible components:

- `benchmark/rules.json`: four normalized construction-safety rules: PPE hard hat, fall harness, guardrail edge, and struck-by equipment.
- `benchmark/prompts/`: prompt templates for normalized model adapters.
- `benchmark/splits/pilot_manifest.csv`: 163 pilot image-rule pairs.
- `benchmark/splits/scaleup_candidate_manifest.csv`: 588 scale-up candidate image-rule pairs.
- `benchmark/annotations/`: pilot and scale-up model-assisted annotation layers, annotation templates, human-audit protocol, AI audit passes, returned adjudication, and final audit labels.
- `benchmark/interventions/`: 948 frozen pilot intervention specifications.
- Frozen model outputs: `results/frozen_model_outputs/`. These include deterministic baselines and local Florence-2 grounding outputs.
- `results/intervention_outputs/`: Florence-2 outputs after targeted and matched-random occlusions.
- `results/tables/`: aggregate score tables, per-example score files, slice diagnostics, intervention summaries, human-audit status, and final-audit score tables.

- `paper/neurips/figures/`: generated compact pipeline, audit-dashboard, intervention-effect, and rule-slice figures.
- `results/release_manifest.json`: SHA-256 manifest for checked artifacts.

## A.2 Benchmark Manifests

The pilot manifest contains 163 examples from the ConstructionSite test split. It was designed for early pipeline validation and includes a relatively balanced set for PPE, fall, and compliant scenes, with struck-by risk underrepresented because that source class is scarce.

Table 8: Pilot manifest class counts.

| Class          | Count |
|----------------|-------|
| PPE violation  | 50    |
| Compliant      | 50    |
| Fall hazard    | 50    |
| Struck-by risk | 13    |
| Total          | 163   |

The scale-up candidate manifest contains 588 image-rule pairs selected from 3,004 scanned test-split source rows with seed 20260719. The target was 250 examples per class where available, but fall-hazard and struck-by-risk cases remained limited by source data availability.

Table 9: Scale-up candidate manifest class counts.

| Class          | Count |
|----------------|-------|
| Compliant      | 250   |
| Fall Hazard    | 75    |
| Ppe Violation  | 250   |
| Struck By Risk | 13    |
| Total          | 588   |

Table 10: Scale-up candidate rule counts.

| Rule                | Count |
|---------------------|-------|
| PPE hard hat        | 303   |
| Guardrail edge      | 110   |
| Struck-by equipment | 106   |
| Fall harness        | 69    |
| Total               | 588   |

## A.3 Annotation Layers

Two model-assisted annotation files have been completed. They are generated from ConstructionSite metadata, captions, source rule-violation records, and available target boxes. They are

useful for debugging the benchmark and prioritizing review, but they are not independent human or domain-expert ground truth.

Table 11: Completed model-assisted annotation status.

| Split    | Rows | Compliant | Violation | Ambiguous yes | Ambiguous no |
|----------|------|-----------|-----------|---------------|--------------|
| Pilot    | 163  | 50        | 113       | 94            | 69           |
| Scale-up | 588  | 250       | 338       | 415           | 173          |

The repository also contains `human_audit_batch_001`, a balanced 120-row audit queue prioritized from scale-up Florence/bootstrap disagreements and weak evidence-overlap cases. It has 30 rows per rule. Among these 120 rows, 118 are Florence/bootstrap answer disagreements, 82 have nonambiguous bootstrap labels, 83 are marked as rule-applicable, 59 have low evidence IoU, and one has missing predicted evidence. The current annotator fields are populated by two independent AI-generated audit passes: annotator A uses a strict compliance-officer persona and annotator B uses a pragmatic site-superintendent persona. A returned adjudication package resolves all 12 A/B answer-label disagreements. The resulting final-label layer is complete, but it should be described with this provenance rather than as unqualified human ground truth.

Table 12: Human audit queue status.

| Metric                               | Value  |
|--------------------------------------|--------|
| Rows prepared                        | 120    |
| Rows per rule                        | 30     |
| Florence/bootstrap disagreements     | 118    |
| AI annotator 1 completion            | 100.0% |
| AI annotator 2 completion            | 100.0% |
| Raw answer agreement                 | 90.0%  |
| A/B disagreements                    | 12     |
| Disagreement adjudication completion | 100.0% |
| Final label coverage                 | 100.0% |

The final-label layer contains 83 violation, 31 compliant, and 6 uncertain labels. It combines 108 A/B consensus labels with 12 returned adjudication decisions. The 12 adjudicated disagreements are concentrated in rule-scope and visibility-threshold cases: six `guardrail_edge` rows involving graded or sloped embankments, four `struck_by_equipment` rows involving static-frame proximity or swing-radius uncertainty, one `ppe_hard_hat` row with borderline headwear visibility, and one `fall_harness` row where the annotators differed on whether an elevated deck pour implied fall-protection need without a visible edge.

Table 13: Audit-final answer scores against the 120-row final-label layer.

| Source               | N   | Accuracy | Macro-F1 |
|----------------------|-----|----------|----------|
| Annotation bootstrap | 120 | 78.3%    | 50.5%    |
| Florence grounding   | 120 | 18.3%    | 12.5%    |
| AI annotator 1       | 120 | 93.3%    | 85.2%    |
| AI annotator 2       | 120 | 95.8%    | 90.5%    |

## A.4 Model Outputs and Answer-Level Scores

The current model-output layer contains deterministic adapters and the first real local image-grounded adapter:

- **Annotation bootstrap:** echoes the model-assisted label and evidence.
- **Manifest seed:** echoes the weak manifest label.
- **Majority violation:** always predicts violation with no image evidence.
- **Caption keyword:** uses only source captions and rule-specific keywords.
- **Florence grounding:** runs `microsoft/Florence-2-base-ft` through the companion local grounding pipeline, then maps detections through deterministic geometry.

Table 14: Pilot model-output scores against model-assisted bootstrap labels.

| Adapter              | Accuracy | Macro-F1 | Evidence present | Nonambig. acc. |
|----------------------|----------|----------|------------------|----------------|
| Annotation bootstrap | 100.0%   | 100.0%   | 98.8%            | 100.0%         |
| Florence grounding   | 41.7%    | 41.7%    | 96.9%            | 37.7%          |
| Manifest seed        | 100.0%   | 100.0%   | 0.0%             | 100.0%         |
| Majority violation   | 69.3%    | 40.9%    | 0.0%             | 72.5%          |

Table 15: Scale-up model-output scores against model-assisted bootstrap labels.

| Adapter              | Accuracy | Macro-F1 | Evidence present | Nonambig. acc. |
|----------------------|----------|----------|------------------|----------------|
| Annotation bootstrap | 100.0%   | 100.0%   | 57.3%            | 100.0%         |
| Florence grounding   | 39.3%    | 38.6%    | 98.8%            | 28.3%          |
| Manifest seed        | 100.0%   | 100.0%   | 0.0%             | 100.0%         |
| Majority violation   | 57.5%    | 36.5%    | 0.0%             | 82.7%          |
| Caption keyword      | 19.7%    | 26.6%    | 0.0%             | 8.1%           |

The Florence grounding adapter produced 163 pilot outputs and 588 scale-up outputs with zero invalid normalized answers. On the pilot split it predicted 111 compliant and 52 violation answers. On scale-up it predicted 399 compliant and 189 violation answers. The low answer accuracy relative to the bootstrap labels is not hidden; it is evidence that the current local grounding proxy is not a strong safety VQA model and should be treated as a diagnostic baseline.

## A.5 Scale-Up Florence Slice Diagnostics

The slice analysis shows where the current local Florence pipeline fails most strongly. The scale-up result is weakest for PPE hard-hat and struck-by equipment examples. It still predicts evidence boxes for almost every example, which means answer failure and evidence availability are distinct failure modes.

Additional slice files are stored as `pilot_florence_grounding_slices.csv` and `scaleup_florence_grounding_slices.csv`. They include overall, rule-level, ambiguity-level, and rule-applicability-level rows with accuracy, invalid-rate, evidence-presence, reference-evidence, mean IoU, and centroid-drift fields.

Table 16: Scale-up Florence rule slices.

| Rule                | N   | Accuracy | Evidence present | Mean IoU |
|---------------------|-----|----------|------------------|----------|
| Fall harness        | 69  | 59.4%    | 98.6%            | 0.130    |
| Guardrail edge      | 110 | 63.6%    | 96.4%            | 0.171    |
| PPE hard hat        | 303 | 30.7%    | 99.3%            | 0.030    |
| Struck-by equipment | 106 | 25.5%    | 100.0%           | 0.033    |

## A.6 Intervention Specifications and Effects

The frozen pilot intervention file contains 948 occlusions: 158 targeted rule-evidence masks and 790 matched-random controls. There are five matched-random controls per targeted mask, using the same clipped mask width and height as the corresponding target.

Table 17: Pilot intervention specification counts.

| Intervention type        | Count |
|--------------------------|-------|
| Targeted Occlusion       | 158   |
| Matched Random Occlusion | 790   |
| Total                    | 948   |

The Florence intervention run completed all 948 masked-image reruns using beam decoding and black-box masks. It produced 649 compliant and 299 violation outputs. The intervention results show the core faithfulness pattern: targeted evidence removal changes answers and evidence locations more than same-size random-location masking.

Table 18: Pilot Florence intervention effects.

| Metric                 | Targeted | Matched random | Paired diff. |
|------------------------|----------|----------------|--------------|
| Answer flip rate       | 39.2%    | 9.2%           | 30.0%        |
| Centroid drift         | 0.188    | 0.029          | 0.160        |
| Evidence disappearance | 5.1%     | 3.7%           | —            |

The paired answer-flip difference is 30.0 percentage points with a bootstrap 95% interval of 22.3 to 37.7 percentage points. The paired centroid-drift difference is 0.160 with a bootstrap 95% interval of 0.133 to 0.187. Evidence disappearance occurred in 5.1% of targeted rows and 3.7% of matched-random rows, so disappearance is tracked separately from ordinary evidence relocation.

## A.7 Validation and Test Results

The command `python./experiments/validate_all.py` currently validates:

- benchmark rules and prompt templates;
- pilot and scale-up model-assisted annotations;
- the 120-row human-audit batch, returned adjudication files, and final-label layer;
- pilot and scale-up normalized model-output files;

- the 948-row Florence intervention output file;
- Phase 4 intervention specifications and generated score tables;
- the NeurIPS paper scaffold, required generated tables, and generated figures;
- the 588-row scale-up candidate manifest;
- 37 Python tests across `analysis/tests` and `tests`.

The latest validation run passed all checks. The release manifest hashes benchmark data, experiment scripts, paper files, generated figures, frozen model outputs, intervention outputs, result tables, source code, and tests.

## A.8 Regeneration Commands

The following commands regenerate the current major artifacts. Florence commands require the companion local XAI pilot environment at `../Explainable-AI-Mustafa-Abdallah/pilot/.venv`.

```
python .\experiments\build_pilot_manifest.py
python .\experiments\build_model_assisted_annotations.py
python .\experiments\build_scaleup_candidate_manifest.py
python .\experiments\build_scaleup_annotations.py

python .\experiments\run_model_harness.py --adapter annotation_bootstrap
python .\experiments\run_model_harness.py --adapter manifest_seed
python .\experiments\run_model_harness.py --adapter majority_violation
python .\experiments\run_model_harness.py --adapter caption_keyword

$python = '..\Explainable-AI-Mustafa-Abdallah\pilot\.venv\Scripts\python.exe'
& $python .\experiments\run_florence_grounding_outputs.py '
--manifest .\benchmark\splits\pilot_manifest.csv '
--output-stem pilot_florence_grounding
& $python .\experiments\run_florence_grounding_outputs.py '
--manifest .\benchmark\splits\scaleup_candidate_manifest.csv '
--output-stem scaleup_florence_grounding

python .\experiments\build_intervention_specs.py
& $python .\experiments\run_florence_intervention_outputs.py '
--interventions .\benchmark\interventions\pilot_interventions.jsonl '
--output-stem pilot_florence_interventions

python .\experiments\score_model_outputs.py '
--model-output .\results\frozen_model_outputs\pilot_florence_grounding.jsonl '
--name pilot_florence_grounding
python .\experiments\score_model_outputs.py '
--model-output .\results\frozen_model_outputs\scaleup_florence_grounding.jsonl '
--name scaleup_florence_grounding '
--annotations .\benchmark\annotations\scaleup_model_assisted_annotations.jsonl '
--manifest .\benchmark\splits\scaleup_candidate_manifest.csv
python .\experiments\score_intervention_outputs.py
python .\experiments\analyze_score_slices.py '
--per-example .\results\tables\scaleup_florence_grounding_per_example_scores.csv '
--name scaleup_florence_grounding
python .\experiments\build_human_audit_batch.py
python .\experiments\ingest_human_audit_passes.py '
--package "D:\My Projects\human-audit-batch-001-annotation\final_package"
$adjPackage = "D:\My Projects\human-audit-batch-001-annotation\adjudication\" + '
"human_audit_batch_001_adjudication_package"
python .\experiments\ingest_human_audit_adjudication.py '
--package $adjPackage
python .\experiments\score_human_audit_status.py
python .\experiments\score_human_audit_final_labels.py
```

```
python .\experiments\make_paper_tables.py
python .\experiments\make_paper_figures.py
python .\experiments\build_release_manifest.py
python .\experiments\validate_all.py
```

## A.9 Current Limits Before NeurIPS-Style Claims

The implemented results are sufficient for a compact, reproducible audit study. They are not yet sufficient for broad model-ranking claims. The remaining required work is independent human/domain replacement or verification of the AI-pass audit layer, additional instruction-following VLM adapters, larger balanced splits, and final review under the official target conference template.